

Title:**Enhancing Placement Readiness Through AR/VR-Based Mock Interview Simulation:
Development, Evaluation, and Educational Implications****Authors:**

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Abstract:

With today's competitive employment marketplace, educational credentials alone are no longer enough to guarantee effective employment. Employers are more and more holding greater value on so-called "soft skills" like communication effectiveness, poise under pressure, flexibility, and psychological preparedness—particularly when interviewing for placement. Conventional interview preparation strategies, e.g., reading books or conducting mock interviews with fellow students, cannot adequately simulate the high-stakes, engaging atmosphere of an actual interview. In order to fill this important gap, our study proposes a virtual mock interview simulation platform developed through Unity Hub (version 6), utilizing engaging AR/VR technology to improve readiness in students and employability seekers.

The simulation is meant to closely replicate real-life formal interview environments through the use of scripted audiovisual scenes, authentic environments, and behavioural cues. The simulation exposes users to both behavioural and situational interview questions, mimicking pressure situations that prompt genuine user responses. In contrast to traditional mock interviews, the virtual environment facilitates multiple practice sessions in a psychologically secure and reproducible setting, allowing users to identify and improve their weaknesses over time without external criticism.

Technically, the system uses a modular design to accommodate scalability in the future. The core elements are immersive 3D settings, animated interviewee avatars, and synchronous audio feedback. Optimization is done for VR head-mounted display, although future versions plan for multiplatform implementation with desktop and mobile compatibility. A pilot study with four undergraduate participants was conducted to test the effectiveness of the system. Pre- and post-simulation feedback showed remarkable gains in reported confidence, verbal fluency, and preparedness to handle actual interviews

Educationally, the instrument provides a pedagogically engaging learning experience that engages cognitive processing and emotional conditioning. The authenticity of the simulated interview context lowers performance stress by familiarizing users with real-life scenarios ahead of time. Furthermore, the game-like nature of the experience boosts motivation and long-term retention of soft skills. Apart from its independent application, the system is well positioned for incorporation into university career development plans, especially in those that do not have access to high-quality placement materials.

In spite of its promise, some limitations in the prototype right now are that the system fails to provide dynamic feedback or real-time evaluation metrics. Also, the interviewer avatars have scripted logic, which restricts adaptability. Enhancements in the future will include Natural Language Processing (NLP) so that responsive, AI-powered interviewer bots can adapt according to user responses. Real-time analysis and feedback like eye contact analysis, voice modulation scores, and tracking of behavioural cues are also in the pipeline.

Finally, this paper offers a step towards democratizing access to high-quality, interactive interview preparation tools. By combining immersive technology with pedagogical principles, we hope to develop scalable solutions that empower students in various educational institutions. With ongoing innovation and the inclusion of AI, the platform can potentially revolutionize placement readiness training globally.

1. Introduction

Job interviews are perhaps the most tense and important milestones for students making the leap from school to professional settings. Although academic curricula have adjusted to include technical competency, they sometimes lack the systematic preparation of students for actual hiring procedures.

Classic mock interviews are plagued by numerous challenges: non-scalability, variability of feedback, threatening formats, and reliance on human assessors. Additionally, these arrangements are logistically challenging and not possible to scale to meet hundreds or thousands of students at a time.

The millennial and Gen Z learner population, as digital natives, excels in interactive and visually oriented environments. Intense needs exist to rethink interview training through technology-enabled, immersive learning frameworks.

This article presents a new AR/VR-fuelled simulation system that provides an interactive mock interview experience. In contrast to in-person configurations, the system offers a psychologically neutral, exciting, and reproducible space in which students can practice and hone their soft skills. The system was built within Unity and combines realistic graphics, spatial audio, and ordered question sequences. It is compatible with VR and desktop environments and can be easily deployed at scale.

2. Literature Review

that immersive technology can improve learning outcomes by promoting engagement, retention and transfer of skills. Many studies have established that AR/VR can positively influence learners' engagement, retention, and application of skills.

Jensen and Konradsen (2018) reported similar findings related to the use of VR in experiential learning, where learner retention and the level of conceptual understanding continued to grow due to the use of immersive environments. The results from the PwC study (2020) indicated that learners who were trained with VR were 40% more confident in their ability to apply skills compared to those trained in a traditional classroom. Moreover, learners who were trained with VR found they also had

a greater emotional connection to the material being trained on, which appeared to draw out greater levels of engagement.

Chugh and Ruhi (2021) undertook a study of the increased incidence of virtual interviews during the pandemic and the impact this had on students' perceived preparedness. Students reported greater comfort and confidence in the virtual interviews compared to face-to-face interviews, as the researchers found that increased exposure to virtual interviews assisted in the reduction of performance anxiety; and supported reusable communication fluency for the students they studied.

Al-Shammari et al. (2022) offered up a holistic approach for evaluating the effectiveness of virtual training, and noted specific components of immersive realism, feedback components, and learner flexibility were important in their evaluation. Kaur and Pathak (2023) overall analysed AR/VR applications which positively influenced retention of knowledge and participation in the classroom and which would be further enhanced with traditional learning approaches

3. Problem Statement

The main constraints with traditional mock interviews are:

- Resources (the need for experienced interviewers and structured environments)
- The difficulty level of the interview questions, feedback and engagement are not standardized
- Performance anxiety in face-to-face settings
- Students in rural or under-privileged communities are not able to access practice interviews

Our system alleviates these problems by creating an immersive, scalable, and repeatable experience that reflects the real-world dynamic of interviews through AR/VR technology. It provides equitable opportunities to interview training and improves student preparedness with modern tools.

4. System Design and Development

The mock interview system was established within Unity Hub (version 6), as we were able to use the rendering software and take advantage of VR and non-VR settings. The design has three components:

4.1 Virtual Interview Room

To create a realistic corporate interview space in the study, we designed a professional-looking virtual space that included seating arrangements, lighting, and virtual desk arrangements. All of these features were designed with high levels of psychological realism.

4.2 Interview Avatars:

The program has a virtual Human Resource avatar (the interviewer) and a student avatar (the interviewee). The current version uses pre-scripted questions, however, future versions will utilize AI-powered dynamic conversations with students.

4.3 Audio Integration:

The AudioSource component of Unity was used to embed spatial audio for realistic sound perception. Questions were posed through clear, natural-sounding speech, and interview cues were timed contextually.

4.4 UI and Interaction Flow

The interface was designed to be minimalist. Users can navigate the simulation using a set of basic prompts. The interaction follows a linear interview format where the virtual HR presents a sequence of questions, and the student responds verbally or via text in future updates.

4.5 Device Compatibility

The system was tested on VR headsets and desktop computers to ensure accessibility. For users without access to VR gear, the desktop experience offers an engaging 3D simulation.

5. Methodology

To validate the system, we conducted a pilot test involving four student participants who contributed to its development. Each student underwent a 15-minute interview simulation session. No formal psychological scales were used; however, structured feedback was collected.

5.1 Feedback Questions Included:

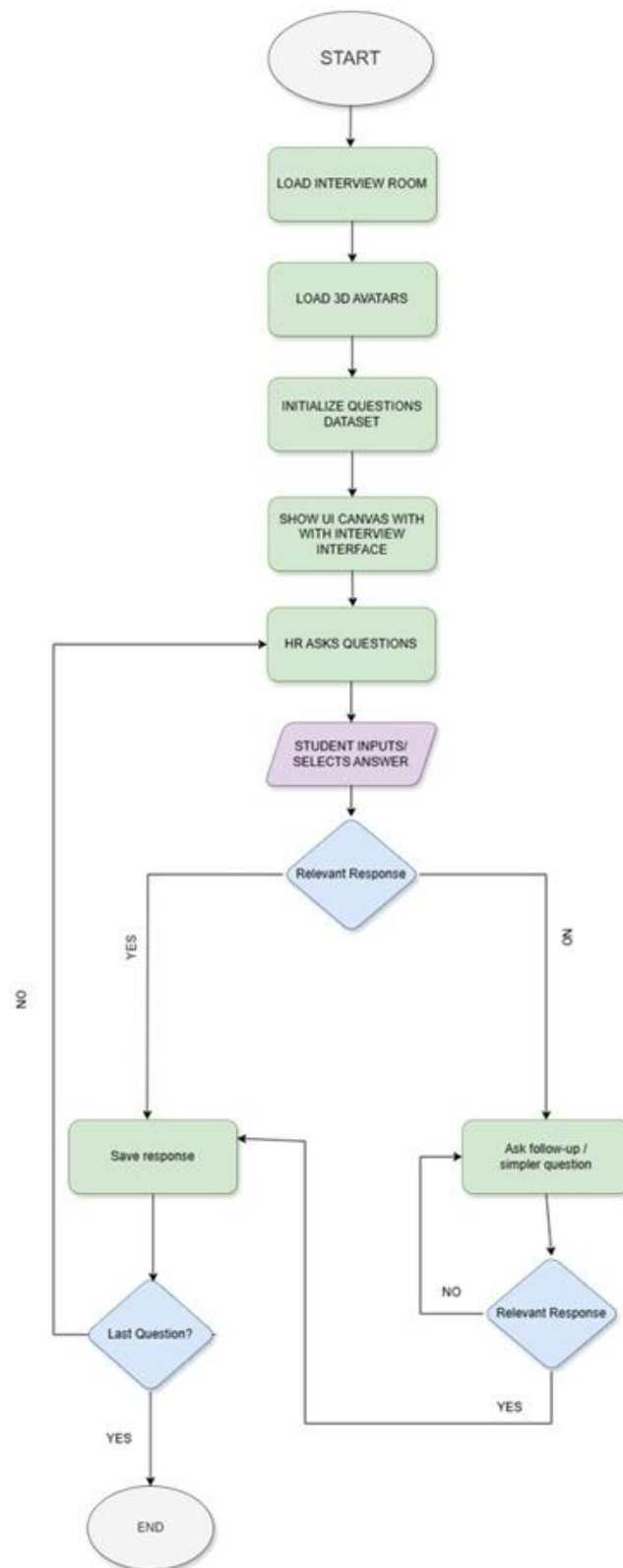
- How realistic did the environment feel?
- Did the simulation reduce your anxiety about interviews?
- How confident do you feel after completing this session?
- What improvements would you make to the system?

5.2 Observations:

All participants reported the following:

- An increase in self-reported confidence
- A stronger sense of preparedness
- Appreciation for the immersive nature of the experience
- Requests for dynamic question sets and AI interviewer support

5.3 Flowchart:



6. Results and Discussion

The feedback from the pilot was positive and encouraging. All participant groups reported that the system enabled them to practice in a realistic and supportive environment. The participants reported feeling less anxious because there was no human judge involved, which freed them to plan their answers before submitting, rather than worrying about how their answer would be judged before going to the next question.

Our findings are in line with the findings of PwC (2020) and Chugh and Ruhi (2021), who noted that virtual roleplays help develop learner confidence, and since our system supports the conclusions from the two studies, it highlights the potential of our system to have an impact.

Nevertheless, we ran into challenges with the following:

- Limited interactivity (just scripted Qs)
- Inadequately small sample size
- Developer-participant bias
- Lack of standardized measurement tools

Nevertheless, we have established a strong fundamental infrastructure for future development.

7. Future Enhancements

We plan on having many upgrades to the tool. These are:

- **AI-Powered Avatars:** To have dynamic conversations based on individual responses
- **Natural Language Processing:** For spoken inputs and real-time feedback
- **Progress Tracking:** Session scores and analytics on performance
- **Gamification:** Leaderboards, badges, and time challenges
- **Question Bank Extension:** Questions sorted relevant to industries and roles
- **Multilingual:** Reach a larger demographic
- **Mobile:** More opportunity for accessibility across platforms

8. Educational Implications

This system could be embedded within the academic curriculum to change the placement roads. It could be:

- A mandatory pre-placement training module
- A self-directed career readiness course
- An intervention resource for below threshold students
- A flip-able plan for institutions dealing with high volumes of students

The immersive format allows students to be engaged and the replayability allows for mastery through practice, which standard mock interview formats cannot replicate.

9. Conclusion

This paper has presented an innovative AR/VR-based simulation tool that addresses important issues with traditional interview preparation. Utilising immersive design and realistic audio-visual interaction, the system is designed to increase student confidence and reduce the psychological barriers to real interviews. Initial findings are optimistic, align with worldwide findings of VR in education, and provide a basis for upgrading the system to include AI and mobile access. With these (and other) enhancements, the system has the potential to become a common tool in global placement training programs.

10. Author Biographies:

Aahilahmed Bakhtiyar Nadaf.

Aahil is an undergraduate engineering student with a keen interest in AR/VR development, automation systems, and the intersection of immersive tech and education. He contributed significantly to the conceptualization, design, and implementation of the mock interview simulation system. Aahil also has experience in user-centered design and system testing, making sure the application delivered both realism and usability. With a strong background in graphics and system workflows, he continues to explore projects that merge creativity, utility, and emerging technologies.

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Purva Suraj Shinde.

Purva is an engineering undergraduate passionate about emerging educational technologies and human-computer interaction. She was instrumental in designing the user experience, conducting user testing, and documenting the methodology for the AR/VR-based mock interview system. Her focus on usability and clarity helped shape the immersive interface and ensured accessibility across devices. She aims to work on projects that drive innovation in education and learning environments through technology.

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Riya Vijay Khulpe.

Riya is an undergraduate student with a strong foundation in research, literature review, and system documentation. For this project, she led the literature synthesis and benchmarking of existing AR/VR solutions in education. Her contributions helped validate the project's core direction and align it with recent academic findings. She is interested in the educational applications of immersive media and continues to explore interdisciplinary projects that combine tech and pedagogy.

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Shreyas Krushnant Chougale.

Shreyas is an undergraduate student with academic interests in software systems. He provided minor support in technical aspects during the initial stages of the project. His involvement included contributions during brainstorming sessions and reviews of early prototypes. He looks forward to expanding his involvement in future research-based projects and developing his expertise in system implementation.

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